Surface Edits, Real Effects: How Whitespace, Casing, and JSON Wrapping Change LLM Outputs Across Scale

NeuriCo

Abstract

Prompt templates often differ in small ways that look harmless: a trailing space, a template written in capitals, missing punctuation, or an input wrapped in JSON. We ask whether such meaning-preserving edits change what instruction-tuned language models output, and whether larger models are less affected. We apply single, atomic surface edits to chat-formatted prompts and run ten open-weight models (QWEN2.5 0.5B–14B, LLAMA-3.X 1B/8B, GEMMA-2 9B) with greedy decoding on about 235k generations. We measure classification accuracy, response length, instruction compliance on IFEVAL, and extraction F1. A replicate run and placebo edits measure the noise floor of the inference stack. Across 40 model×task cells, accuracy spread over ten variants has a median of 5.8 points (range 1.0–28) and exceeds a permutation null in 27 cells. Sensitivity falls with scale (-9.5 points of spread per tenfold increase in parameters, $\rho = -0.87$). The decline is not log-linear, though: it plateaus at about 4 points between 7B and 14B. ALL-CAPS templates are the most harmful edit for small models (up to -27 points) and still lower IFEVAL compliance by 3.1–4.4 points at 7–14B. Two common beliefs do not hold: trailing whitespace does not lengthen outputs (length ratios 0.94–1.07), and JSON-structured prompts do not give a uniform 10–15 point extraction gain (effects range from -17 to $+26$ F1 depending on task and model). Which edits help or hurt varies by model, even between the 16-bit and 4-bit versions of the same weights. We recommend evaluating small models over several formats and avoiding ALL-CAPS templates at any size.

1 Introduction

Every prompt carries formatting choices that nobody thinks of as part of the experiment. A benchmark author writes the instruction in Title case, a developer pastes a template that ends in a newline, and a product team writes the system rules in capitals "for emphasis." These choices do not change what the prompt means. **Do they change what the model outputs?**

The question matters for two groups. For people who evaluate models, a format effect of a few points is as large as many reported gaps between models or methods. A comparison between two prompts or two models is then partly a comparison between two formats. For people who deploy models, a format effect means a production prompt can break after an edit that no code review would flag. Folk advice about such effects is common: trailing whitespace makes answers longer, capitals make models obey less, and JSON prompts make extraction more accurate. These claims are rarely measured.

Prior work has shown that prompt format matters. Sclar et al. [2024] found large accuracy spreads over joint choices of separators, casing, and spacing in few-shot prompts, and found that scale does not remove them. Mizrahi et al. [2024] argued that single-prompt evaluation is unreliable and that benchmarks should report results over many paraphrases. However, three questions remain

39th Conference on Neural Information Processing Systems (NeurIPS 2025).

open. First, most studies pool many format choices into a single spread, so they cannot say *which* edit is responsible. Second, they mostly use completion-style prompts, whereas deployed models receive prompts wrapped in a chat template, which can absorb or strip some edits. Third, they measure accuracy only, and the folk claims about length, compliance, and structured prompts remain untested. A further issue is that LLM inference is not perfectly deterministic, even under greedy decoding. Without a measured noise floor, small format effects cannot be separated from noise in the inference stack.

We run a controlled study of *atomic* surface edits on chat-formatted prompts. We change one aspect of the template at a time (trailing whitespace, double spacing, line breaks, casing, punctuation, separators, or JSON wrapping) and keep the inserted data byte-identical. We test five preregistered hypotheses (section 3) on ten open-weight instruction-tuned models from three families, spanning 0.49B to 14.7B parameters. We measure four outcomes: classification accuracy on four tasks, response length and instruction compliance on IFEVAL Zhou et al. [2023], and extraction F1 on SQUAD and CONLL. All comparisons are paired at the item level. Every effect is compared against a permutation null and against the inference stack's own run-to-run noise, which we measure with a full replicate run and with "placebo" edits that the chat template removes before the model sees them.

What do we find? Surface edits matter, but not uniformly. The median accuracy spread across ten variants is 5.8 points over 40 model$\times$task cells, and models of 1.2B parameters or fewer show spreads of 7–28 points. Sensitivity drops steeply up to about 1.5B parameters and then levels off at about 4 points of spread and 4% of predictions flipped. QWEN2.5-7B and QWEN2.5-14B are indistinguishable. ALL-CAPS is the most harmful edit for small models (up to -27 points on SST-2) and still costs 3.1–4.4 points of IFEVAL compliance at 7–14B (significant at 7B and 14B). Two folk claims fail. Trailing whitespace does not lengthen answers (length ratios 0.94–1.07, none significant), and JSON-structured prompts do not give a uniform 10–15 point gain: wrapping the *input* in JSON mostly hurts small models, and requesting JSON *output* changes F1 by -14 to $+26$ points depending on task and model. Finally, the direction of most edit effects is model-specific. Edit effects correlate only at $r = 0.19$–0.28 between the 16-bit and 4-bit AWQ versions of the same model, compared with $r = 0.996$ between two runs of the same model.

Our contributions are:

- We design a controlled protocol that attributes format sensitivity to individual surface edits on chat-formatted prompts, with paired tests, permutation nulls, and an explicitly measured inference-noise floor.
- We conduct experiments on 10 models (0.49B–14.7B, three families, 16-bit and 4-bit weights) and about 235k generations, covering classification, verbosity, instruction compliance, and extraction.
- We give quantitative verdicts on five common hypotheses. Two are partially supported (accuracy spread, ALL-CAPS compliance penalty), one is directionally supported but not log-linear (scale), and two are refuted (trailing whitespace lengthens outputs; JSON prompts improve extraction).
- We identify mechanisms behind large effects, including label-vocabulary drift, degenerate abstention, and over-structured answers, and derive practical recommendations for evaluation and prompt design.

Section 2 reviews related work, section 3 describes the setup, section 4 presents results, and section 5 discusses implications and limitations.

2 Related Work

Prompt format sensitivity. Early work on in-context learning showed that accuracy depends on the choice and order of demonstrations Zhao et al. [2021], Lu et al. [2022], and that models can do as well with irrelevant or misleading instructions as with good ones Webson and Pavlick [2022]. Sclar et al. [2024] isolated *formatting*: by jointly varying separators, spacing, and casing in few-shot completion prompts, they found spreads of tens of accuracy points that did not vanish with scale or instruction tuning. Voronov et al. [2024] showed that template choice can outweigh the gains of proposed in-context learning methods. Salinas and Morstatter [2024] found that small changes, such as adding a space or changing the requested output format, flip a noticeable fraction of predictions in chat models, and He et al. [2024] found sizeable differences between plain-text, Markdown, JSON, and YAML prompt templates for GPT models. Robustness benchmarks such as PromptBench Zhu

et al. [2023] study adversarial character- and word-level perturbations. Unlike these studies, we apply one atomic edit at a time to a fixed chat-formatted template, which lets us attribute each effect to a specific edit. We also measure length and compliance in addition to accuracy, and we compare every effect with a measured noise floor.

Multi-prompt evaluation. Given this sensitivity, Mizrahi et al. [2024] argued that LLM benchmarks should report performance over a distribution of instruction paraphrases, and Polo et al. [2024] proposed estimating that distribution efficiently. Evaluation harnesses also stress that small template details affect reproducibility Biderman et al. [2024]. Our results support this agenda and refine it: the need for multi-format evaluation is largest below about 1.5B parameters, while at 7–14B single-edit effects on classification are mostly near noise level.

Instruction following and structured output. IFEVAL Zhou et al. [2023] measures compliance with verifiable instructions such as word counts, required keywords, and letter case. We use its checkers to test whether casing changes compliance. Requesting structured output is another common practice. Tam et al. [2024] found that strict format constraints such as JSON can reduce reasoning accuracy. We separate two things that are often conflated: wrapping the *input* in JSON and requesting JSON *output*. We find that the second has much larger and task-dependent effects than the first.

Nondeterminism in LLM inference. Greedy decoding is often treated as deterministic, but floating-point results in batched inference depend on batch composition, so outputs can change between runs Atil et al. [2024], He and Thinking Machines Lab [2025]. We quantify this directly for our stack (VLLM Kwon et al. [2023]). Batch-dependent noise is negligible for short classification outputs but moves IFEVAL compliance by up to about 3 points, which sets a lower limit on the effects we can detect for long generations.

3 Experimental Setup

Our main question is whether a meaning-preserving edit to a prompt's surface form changes a model's output, and whether this sensitivity shrinks with model size. We hold the semantic content of every prompt fixed, apply one edit at a time, and compare each edited prompt with the clean `base` prompt on the same item.

Hypotheses. We test five specific claims, fixed before running experiments:

- **H1**: meaning-preserving formatting changes produce a 3–8 point accuracy spread on classification.
- **H2**: trailing whitespace in the user message increases output length by 15–25%.
- **H3**: ALL-CAPS instructions reduce compliance by 5–12 points in small models, with a negligible effect ($<$2 points) at $\geq$7B.
- **H4**: JSON-structured prompts improve extraction accuracy by 10–15 points at all sizes.
- **H5**: format sensitivity decreases log-linearly with parameter count.

3.1 Models and Decoding

Table 1 lists the ten models. The 4-bit AWQ Lin et al. [2024] QWEN2.5 ladder from 0.5B to 14B Qwen Team [2024] is the cleanest scale comparison, because it holds the model family and quantization fixed. The 16-bit (bf16) 0.5B and 1.5B runs serve as a quantization control, and LLAMA-3.X Grattafiori et al. [2024] and GEMMA-2 Gemma Team [2024] add two more families. All models run locally with VLLM 0.32.0 Kwon et al. [2023] on shared NVIDIA RTX A6000 GPUs. We use greedy decoding (temperature 0, seed 42), each model's default chat template and default system prompt, and a single user turn. The generation limit is 10 tokens for classification, 1024 for IFEVAL, 64 for SQUAD, and 256 for CONLL. Each model receives 24,046 prompts. GEMMA-2-9B did not finish its IFEVAL generations, so experiments on length and compliance cover nine models.

Table 1: Models evaluated. All are instruction-tuned. AWQ = 4-bit activation-aware weight quantization. GEMMA-2-9B has no IFEVAL results.

Short name	Checkpoint	Params	Weights
qwen0.5b	Qwen2.5-0.5B-Instruct	0.49B	bf16
qwen0.5b-awq	Qwen2.5-0.5B-Instruct-AWQ	0.49B	AWQ int4
llama1b	Llama-3.2-1B-Instruct	1.24B	bf16
qwen1.5b	Qwen2.5-1.5B-Instruct	1.54B	bf16
qwen1.5b-awq	Qwen2.5-1.5B-Instruct-AWQ	1.54B	AWQ int4
qwen3b-awq	Qwen2.5-3B-Instruct-AWQ	3.09B	AWQ int4
qwen7b-awq	Qwen2.5-7B-Instruct-AWQ	7.62B	AWQ int4
llama8b-awq	Meta-Llama-3.1-8B-Instruct-AWQ-INT4	8.03B	AWQ int4
gemma9b-awq	gemma-2-9b-it-AWQ-INT4	9.24B	AWQ int4
qwen14b-awq	Qwen2.5-14B-Instruct-AWQ	14.7B	AWQ int4

Table 2: Surface edits applied to the classification template. Every edit preserves the meaning of the prompt and leaves the inserted data unchanged.

Variant	Edit
`base`	clean template
`trailing_space` / `trailing_newline`	append `" "` / `"\n"` to the user message
`double_space`	double every space in the template
`single_line`	replace newlines in the template with spaces
`lowercase`/ `uppercase`	lower-case / UPPER-case the template
`no_punct`	remove `.,:;?!` from the template
`exclaim`	replace `.` with `!`
`sep_dash`	replace field separator `:` with `-`

3.2 Tasks and Surface Edits

Exp. 1: classification (H1, H5). We use SST-2 validation Socher et al. [2013] (n=500), AG NEWS test Zhang et al. [2015] (500), EMOTION test Saravia et al. [2018] (500), and BOOLQ validation Clark et al. [2019] (300, passages of at most 200 words). The template has an instruction, the input in a labelled field (e.g., "`Article: {input}`"), a line stating the answer format, and an answer cue (e.g., "`Topic:`"). Table 2 lists the nine edits. Each edit modifies the *template text only*; the dataset input is inserted verbatim. We take the first label word of the response as the prediction (prefix match, so "sport" maps to *sports*) and count unparseable outputs as wrong. Parse rates are 87–100%.

Exp. 2 and 3: length and compliance (H2, H3). We use all 541 IFEVAL prompts Zhou et al. [2023] under six conditions: `base`, `trailing_space`, `trailing_newline`, `trailing_2newline`, `uppercase`, and `lowercase`. We measure response length in generated tokens. We measure compliance with the official IFEVAL checkers (taken from the lm-evaluation-harness Biderman et al. [2024]), always evaluated against the *original* prompt's instruction arguments. Our main metric is prompt-level strict accuracy. Because upper-casing can change the meaning of instructions about letter case, we also report the effect after excluding prompts with `change_case` instructions.

Exp. 4: extraction (H4). We use SQUAD v1.1 validation Rajpurkar et al. [2016] (n=400; EM and F1 with the official normalization) and CONLL-2003 test Tjong Kim Sang and De Meulder [2003] (300 sentences with at least one entity; micro-F1 over (type, text) pairs). We use a 2×2 design. The *input format* is either labelled plain text or a JSON object with the same fields as keys (`task`, `passage` and `question` for SQUAD or `sentence` for CONLL, and `output_format`); see appendix A. The *output format* asks for either a plain answer or a JSON answer. The semantic content is identical across the four cells. One lenient parser scores all cells: it accepts fenced JSON, string-valued and duplicate-key JSON, entity lists, and plain `TYPE: text` lines. Differences therefore reflect the model's answers rather than the parser.

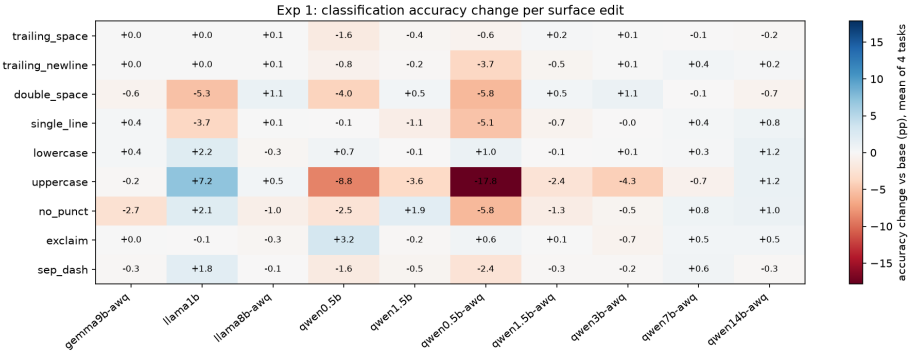


Figure 1: Classification accuracy change (pp) of each surface edit relative to `base`, averaged over SST-2, AG NEWS, EMOTION, and BOOLQ. Red cells mark accuracy losses and blue cells mark gains. ALL-CAPS (`uppercase`) causes the largest losses in small QWEN2.5 models (up to -17.8 on average for qwen0.5b-awq) but *raises* accuracy for llama1b ($+7.2$). Trailing whitespace is a byte-identical placebo for LLAMA-3.X and GEMMA-2 (exactly 0.0 or 0.1) and has small effects elsewhere. Models of 7B parameters or more have $|\Delta| \leq 1.2$ for every edit except `no_punct` on GEMMA-2-9B (-2.7).

3.3 Metrics and Statistics

All comparisons are paired at the item level.

- **Variant vs. base.** We use an exact McNemar test on correct/incorrect pairs, with Holm correction over the nine variants within each model×task, and paired bootstrap 95% confidence intervals (2,000 resamples).
- **Spread.** We define spread as the maximum minus the minimum accuracy across the ten variants. With n=300–500, the max–min of ten noisy estimates is positive even when there is no effect. We therefore compare it to a permutation null that shuffles each item's outcomes across variants (1,000 permutations).
- **Flip rate.** The fraction of items whose predicted label differs from the `base` prediction, averaged over variants.
- **Length.** We use the Wilcoxon signed-rank test on per-item lengths and a bootstrap CI on the ratio of mean lengths.
- **Extraction.** We use a paired bootstrap on per-item F1 differences.
- **Scale.** We fit OLS of per-model sensitivity on $\log_{10}(\text{params})$ and also report Spearman's ρ.

Noise floor. We measure the inference stack's own variability in two ways. First, we rerun all 24,046 prompts for qwen1.5b-awq. Second, the LLAMA-3.X and GEMMA-2 chat templates strip trailing whitespace from the user message (a Jinja `| trim` filter), so for these models the trailing-whitespace variants are byte-identical to `base` after templating. Any difference between them is pure inference noise. These *placebo* conditions come for free in our design.

4 Results

We present results by hypothesis: classification spread (section 4.1), length (section 4.2), compliance (section 4.3), extraction (section 4.4), and scale (section 4.5). We close with the noise floor and the consistency of edit effects across models (section 4.6).

4.1 Surface Edits Shift Classification Accuracy, Mostly in Small Models (H1)

Figure 1 shows the mean accuracy change of each edit, and table 3 shows the resulting spread per model and task.

Format effects are real and often large. Over the 40 model×task cells, the median spread is 5.8 points (IQR 2.4–11.9; range 1.0–28.0). The spread exceeds the permutation null in 27 of 40

Table 3: Accuracy spread (max − min over 10 variants, pp) per model and task. Parentheses: mean spread under the permutation null (no format effect). *: spread exceeds the null at $p < 0.05$. Lowest spread per column in **bold**. Models are sorted by size. Models of 1.2B parameters or fewer show 7–28 pp spreads. At 7–14B, only AG NEWS consistently exceeds the null by a wide margin.

Model	SST-2	AG NEWS	EMOTION	BOOLQ	**Mean**
qwen0.5b	11.0 (2.2)*	24.0 (4.0)*	14.4 (4.1)*	6.7 (4.3)*	14.0
qwen0.5b-awq	28.0 (3.6)*	22.4 (4.4)*	26.4 (4.4)*	8.3 (5.3)*	21.3
llama1b	14.8 (2.7)*	20.0 (3.1)*	7.2 (3.6)*	25.7 (5.4)*	16.9
qwen1.5b	2.2 (1.3)*	20.4 (3.2)*	2.4 (2.9)	2.3 (2.8)	6.8
qwen1.5b-awq	1.2 (1.3)	11.2 (2.8)*	6.0 (3.1)*	3.3 (3.0)	5.4
qwen3b-awq	3.8 (1.5)*	10.6 (2.4)*	8.8 (2.9)*	**1.3** (2.3)	6.1
qwen7b-awq	1.8 (1.2)	9.8 (2.3)*	**2.0** (2.1)	2.0 (1.7)	**3.9**
llama8b-awq	**1.0** (1.3)	10.6 (3.1)*	4.2 (2.5)*	2.7 (2.3)	4.6
gemma9b-awq	**1.0** (0.7)	14.2 (2.4)*	2.8 (2.0)	1.7 (1.3)	4.9
qwen14b-awq	2.6 (1.2)*	**3.8** (2.0)*	5.6 (2.5)*	3.7 (2.4)*	**3.9**

Table 4: IFEVAL results. *Left*: ratio of mean response length (tokens) under each variant to `base`, with 95% bootstrap CI for trailing whitespace. †: placebo, since the chat template trims trailing whitespace. *: Wilcoxon $p < 0.05$. *Right*: prompt-level strict accuracy at `base` and change (pp) under ALL-CAPS (with 95% CI) and lowercase. "excl." removes prompts with letter-case instructions. **Bold**: McNemar $p < 0.05$ (uncorrected). All nine ALL-CAPS estimates are negative.

	Length ratio (variant / base)				**Strict compliance (pp)**			
Model	`trailing_space`	`trailing_newline`	`trailing_2newline`	`uppercase`	`base`	`uppercase` Δ	`lowercase` Δ	`uppercase` Δ excl.
qwen0.5b	0.95 [0.90, 1.00]	0.94 [0.88, 1.00]	0.95 [0.89, 1.01]	1.03	27.2	**−5.7** [−9.8, −1.8]	−3.9	−7.2
qwen0.5b-awq	1.00 [0.95, 1.04]	0.95 [0.90, 1.02]	0.99 [0.93, 1.05]	1.00	27.4	**−6.7** [−10.7, −2.6]	−4.3	−8.3
llama1b†	0.99 [0.96, 1.03]	0.99 [0.96, 1.03]	0.98 [0.95, 1.02]	1.05*	47.9	**−4.4** [−8.1, −0.4]	−0.7	−5.9
qwen1.5b	1.02 [0.96, 1.08]	0.98 [0.92, 1.04]	1.07 [1.01, 1.14]	1.10*	41.4	−3.9 [−7.9, −0.2]	−3.3	−5.9
qwen1.5b-awq	0.96 [0.91, 1.00]	0.97 [0.91, 1.03]	0.99 [0.93, 1.06]	1.01	37.5	−3.7 [−7.4, 0.2]	−3.3	−4.4
qwen3b-awq	0.98 [0.93, 1.04]	1.02 [0.96, 1.09]	0.98 [0.92, 1.05]	1.01*	58.2	−3.3 [−7.6, 0.9]	−3.5	−4.6
qwen7b-awq	1.00 [0.97, 1.02]	1.01 [0.97, 1.05]	1.02 [0.98, 1.06]	1.05*	71.7	**−4.4** [−8.1, −0.9]	−2.8	−4.2
llama8b-awq†	1.00 [1.00, 1.01]	1.00 [1.00, 1.01]	0.99 [0.98, 1.00]	1.03*	74.1	−3.1 [−6.7, 0.6]	−2.2	−2.6
qwen14b-awq	0.98 [0.95, 1.00]	0.99 [0.96, 1.03]	1.00 [0.96, 1.03]	1.02*	77.8	**−3.7** [−7.2, −0.4]	−2.0	−5.0

cells. Of the 360 variant-vs-base tests, 48 are significant after Holm correction. The edits do not contribute equally: `uppercase` accounts for 16 of the 48 significant changes, followed by `no_punct` (9), `single_line` (6), `double_space`, `lowercase`, and `trailing_newline` (4 each), `exclaim` and `sep_dash` (2 each), and `trailing_space` (1). The mean prediction flip rate relative to `base` follows the same order: 17.6% for `uppercase`, 14.7% for `no_punct`, 9.8% for `double_space`, and only 1.6% for `trailing_space`.

ALL-CAPS hurts small models only. The largest single effect is -27 points for qwen0.5b-awq on SST-2. Averaged over tasks, `uppercase` costs -17.8 (qwen0.5b-awq), -8.8 (qwen0.5b), and -2.4 to -4.3 points (QWEN2.5 1.5B–3B). For models of 7B or more the effect is near zero (-0.7 to $+1.2$). The sign is not universal: llama1b *gains* 7.2 points under ALL-CAPS.

AG NEWS drives most of the large-model spread. AG NEWS is the most format-sensitive task, with spread $\geq$9.8 points for every model except qwen14b-awq. Inspecting the outputs, most of this swing comes from *label-vocabulary drift*: under some formats, models answer with labels outside the allowed list, such as "politics" for *world* or "news", and how often they do so depends on the format. For models of 7B or more, spreads on SST-2, EMOTION, and BOOLQ lie within about 2 points of the permutation null.

Verdict on H1: partially supported. Format-induced differences are real and usually significant. The 3–8 point range describes the 1.5B–14B models on average (3.9–6.8 points). Models of 1.2B or fewer reach 14–21 points on average, well above the hypothesized range, and at 7–14B much of the remaining spread outside AG NEWS is at noise level.

4.2 Trailing Whitespace Does Not Lengthen Outputs (H2)

The left half of table 4 reports length ratios on IFEVAL. Trailing-whitespace ratios lie between 0.94 and 1.07 and fall on both sides of 1. None of the 21 QWEN2.5 trailing-whitespace comparisons is significant (smallest Wilcoxon $p = 0.06$). This variation matches the placebo models (0.98–

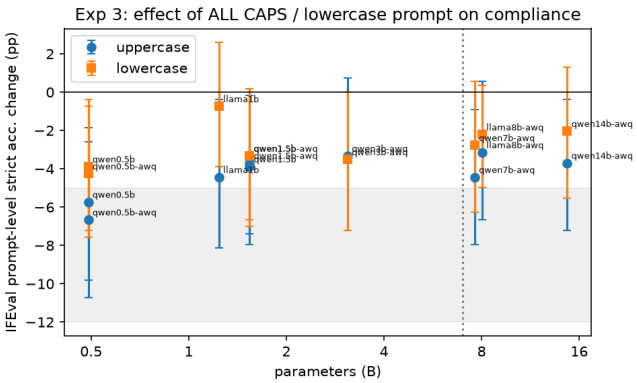


Figure 2: Change in IFEVAL prompt-level strict accuracy under ALL-CAPS (blue) and lowercase (orange) prompts, with 95% bootstrap CIs. The shaded band is the hypothesized 5–12 pp drop for small models, and the dotted line marks 7B. All ALL-CAPS estimates are negative, including at 7–14B, where H3 predicted no effect. Lowercase also hurts, so the cause is non-canonical casing rather than "shouting".

1.00) and the replicate run (0.985), so it is consistent with noise. For LLAMA-3.X and GEMMA-2 the edit cannot have any effect, because their chat templates strip trailing whitespace. In contrast, ALL-CAPS prompts make answers slightly *longer* ($+1$ to $+10\%$, significant for 6 of 9 models).

Verdict on H2: refuted. The largest point estimate is $+7\%$, well short of the hypothesized 15–25%, and no trailing-whitespace effect is significant. A plausible source of the belief is raw completion-mode prompting, where a trailing space changes tokenization at the point where generation starts. In a chat template the whitespace comes before the end-of-turn token, and two of the three families remove it.

4.3 ALL-CAPS Lowers Instruction Compliance at Every Size (H3)

The right half of table 4 and figure 2 show the compliance results. All nine ALL-CAPS point estimates are negative, and five are significant: both 0.5B models, llama1b, qwen7b-awq (-4.4, $p = 0.02$), and qwen14b-awq (-3.7, $p = 0.04$). Excluding prompts whose instructions concern letter case leaves the effects similar or larger (-2.6 to -8.3), so the drop is not an artifact of case-related instructions. Lowercasing also lowers compliance (-0.7 to -4.3), though less than upper-casing. This suggests that the harm comes from non-canonical casing in general rather than from "shouting" specifically. The effect shrinks only slightly with scale: $+1.6$ points per tenfold increase in parameters ($p = 0.028$, $\rho = 0.79$ across nine models; not significant within the 5-model QWEN2.5-AWQ ladder, $p = 0.23$).

Verdict on H3: partially supported. Only the 0.5B models fall in the predicted 5–12 point range. At 7–14B the effect is -3.1 to -4.4 points, which is not negligible: it is significant for two of three models, although it sits close to the 1–3 point inference noise floor (section 4.6).

4.4 JSON Prompts Do Not Uniformly Improve Extraction (H4)

Table 5 separates the two meanings of "JSON-structured prompt".

JSON input mostly hurts small models and has mixed effects on large ones. Wrapping the input in JSON lowers F1 for most models of 1.5B or fewer, by 1.6–13.5 points. At 7–14B the effect depends on the task. On SQUAD it helps llama8b-awq ($+8.5$), qwen7b-awq ($+5.5$), and gemma9b-awq ($+2.3$), and it does not change qwen14b-awq (-0.5). On CONLL the same models mostly lose: llama8b-awq drops 17.4 points and qwen14b-awq drops 4.5. When the sentence is a JSON string value, the LLAMA-3.X models in particular extract fewer entities.

JSON output has large effects whose sign depends on the task. Requesting a JSON answer raises CONLL F1 by 24.8–25.7 points for qwen3b-awq and qwen7b-awq, but lowers their SQUAD F1 by 7.4 and 14.2 points. The two failure modes are visible in the outputs. On CONLL with plain-list

Table 5: Extraction F1 (×100). "Plain": F1 with plain input and plain output. "JSON in": main effect of wrapping the input in JSON, averaged over output formats, with 95% paired bootstrap CI. "JSON out": effect of requesting JSON output with plain input. **Bold**: CI excludes zero and the effect is positive. Requested JSON was valid in 86–100% of responses. No model gains 10–15 points on both tasks.

	SQuAD (span QA)			CoNLL (NER)		
Model	Plain	JSON in Δ	JSON out Δ	Plain	JSON in Δ	JSON out Δ
qwen0.5b	59.7	−4.4 [−6.7, −2.1]	+11.2	17.1	−5.5 [−7.4, −3.7]	−1.1
qwen0.5b-awq	49.9	−13.5 [−16.5, −10.4]	+10.1	10.7	−2.4 [−3.5, −1.2]	+4.4
llama1b	61.6	−5.7 [−8.5, −2.9]	+7.5	21.5	−13.5 [−15.7, −11.2]	+0.1
qwen1.5b	82.4	−1.6 [−3.3, −0.1]	+2.2	26.6	−2.0 [−4.1, 0.0]	+11.4
qwen1.5b-awq	81.4	−1.7 [−3.7, 0.1]	+0.5	26.5	+1.2 [−0.4, 2.8]	−0.6
qwen3b-awq	80.6	−2.0 [−4.1, −0.0]	−7.4	1.4	**+2.5** [0.7, 4.4]	+25.7
qwen7b-awq	82.2	**+5.5** [3.8, 7.4]	−14.2	27.8	−3.4 [−5.4, −1.4]	+24.8
llama8b-awq	64.8	**+8.5** [6.2, 10.9]	−5.9	40.3	−17.4 [−19.8, −14.8]	+12.3
gemma9b-awq	81.4	**+2.3** [1.0, 3.5]	−2.1	**51.4**	+0.8 [−1.0, 2.5]	−0.6
qwen14b-awq	**87.2**	−0.5 [−1.5, 0.7]	+0.9	44.8	−4.5 [−6.5, −2.5]	+6.4

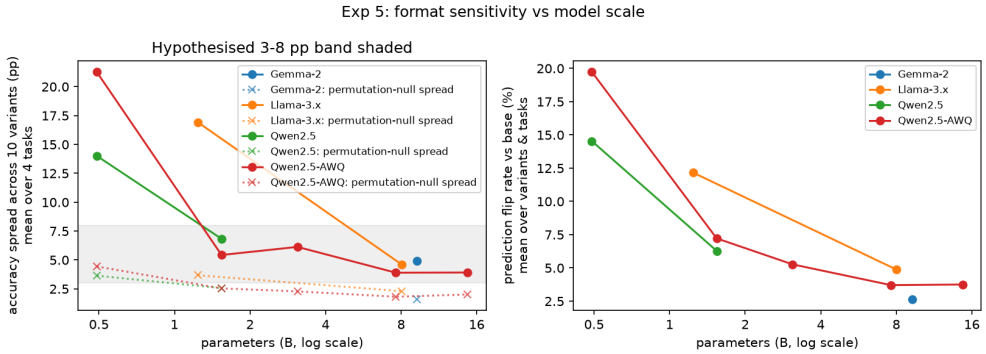


Figure 3: Format sensitivity versus parameter count (log scale). *(Left)*Mean classification spread over four tasks; dotted lines with crosses show the permutation-null spread, and the shaded band is the hypothesized 3–8 pp range. *(Right)*Mean prediction flip rate relative to `base`. Sensitivity drops steeply from 0.5B to 1.5B and then flattens. qwen7b-awq and qwen14b-awq have the same mean spread (3.9 pp) and nearly the same flip rate (3.7% vs. 3.8%).

output, qwen3b-awq answers "NONE" for 97% of sentences (F1 1.4); asked for JSON, it extracts entities (F1 27.1). On SQuAD with JSON output, the Qwen2.5 models put longer, paraphrased spans inside `{"answer": ...}`, which lowers token-level F1.

Verdict on H4: refuted as stated. No model gains 10–15 points from a JSON-structured prompt on both tasks. Structured output helps multi-entity extraction and can hurt short-span QA, so the output format should be chosen and validated per task.

4.5 Sensitivity Falls with Scale, Then Plateaus (H5)

Figure 3 and table 6 relate sensitivity to model size. Across all ten models, spread falls by 9.5 points per tenfold increase in parameters ($p = 0.004$, $\rho = -0.87$) and flip rate by 9.2 points ($p = 0.001$, $\rho = -0.93$). However, the decline is concentrated below 1.5B. Along the Qwen2.5-AWQ ladder, mean spread goes 21.3 → 5.4 → 6.1 → 3.9 → 3.9 points from 0.5B to 14B, and flip rate goes 19.7 → 7.2 → 5.3 → 3.7 → 3.8%. A straight line in log(params) fits this steep drop followed by a floor poorly. Part of the floor is item-level noise (null spreads of 0.7–3.1 points at 7–14B). Part of the decrease may also reflect higher base accuracy (52–63% at the smallest sizes vs. 75–81% at the largest), which leaves fewer items that can flip.

Verdict on H5: directionally supported, but not log-linear. Larger models are less format-sensitive, but the curve flattens at about 4 points of spread and 4% flips from 7B to 14B.

Table 6: OLS slope of per-model sensitivity on $\log_{10}(\text{params})$, i.e., change per tenfold increase in parameters. Negative slopes for spread and flip rate, and positive slopes for (negative) compliance and F1 deltas, mean less sensitivity at larger scale.

Metric	**Models (n)**	**Slope / 10×**	p	**Spearman ρ**
Classification spread	all (10)	−9.5 pp	**0.004**	−0.87
Classification spread	QWEN2.5-AWQ ladder (5)	−10.6 pp	0.084	−0.80
Prediction flip rate	all (10)	−9.2 pp	**0.001**	−0.93
Prediction flip rate	QWEN2.5-AWQ ladder (5)	−10.2 pp	0.051	−0.90
Mean $\lvert\Delta\rvert$ accuracy	all (10)	−2.2 pp	**0.012**	−0.84
ALL-CAPS compliance Δ	with IFEVAL (9)	+1.6 pp	**0.028**	+0.79
Trailing-ws $\lvert \log \rvert$ length ratio	QWEN2.5 (7)	−2.1%	0.061	−0.80
JSON-input F1 Δ	all (10)	+4.1 pp	0.083	+0.60

Table 7: Noise floor of the inference stack. Placebo edits are byte-identical to `base` after chat templating. Short classification outputs are almost unaffected by run-to-run nondeterminism, but long IFEVAL generations often diverge, which moves compliance by up to ∼3 pp.

Check	**Task**	**Identical outputs**	**Outcome difference**
qwen1.5b-awq replicate run	classification	99.7%	max cell $\lvert\Delta$acc$\rvert$ 0.4 pp; edit-effect correlation $r = 0.996$
qwen1.5b-awq replicate run	IFEVAL	76%	length ratio 0.985; strict acc. −0.7 pp
llama1b placebo	classification / IFEVAL	99.6% / 23–31%	max $\lvert\Delta$acc$\rvert$ 0.4 pp / length 0.98–0.99, strict +1.5 to +2.8 pp
llama8b-awq placebo	classification / IFEVAL	99.8–99.9% / 75–76%	max $\lvert\Delta$acc$\rvert$ 0.4 pp / length 0.99–1.00, strict −0.6 to −1.1 pp
gemma9b-awq placebo	classification	99.7%	0.0 pp

4.6 Noise Floor and Idiosyncrasy of Edit Effects

How noisy is greedy decoding? Table 7 shows that greedy decoding in VLLM is not bitwise deterministic across batch compositions. For classification, 99.6–99.9% of outputs are identical across runs, and no cell moves by more than 0.4 points, which is far below the effects in table 3. For long IFEVAL generations, only 23–76% of outputs are identical. This alone moves strict compliance by up to 2.8 points (llama1b placebo), so IFEVAL effects of about 3 points or less cannot be separated from inference noise. Separately, item sampling at $n = 300$–500 produces a spread of 0.7–5.4 points (mean 2.7) even with no format effect, as the permutation nulls show.

Do edit effects transfer across models? Mostly not. Excluding `uppercase`, per-cell edit effects correlate only weakly between the bf16 and 4-bit AWQ versions of the *same* model ($r = 0.28$ at 0.5B, $r = 0.19$ at 1.5B), compared with $r = 0.996$ between two runs of one model. Across other model pairs, correlations range from -0.90 (gemma9b-awq vs. qwen1.5b) to $+0.77$ (llama8b-awq vs. gemma9b-awq), with no clear family structure. Because effects at $\geq$7B are small, these correlations are noisy. Still, apart from the ALL-CAPS penalty, *which* formats help or hurt behaves like model-specific noise, and even quantizing the same weights reshuffles it. Quantization also changes the *magnitude* of sensitivity at small scale: 4-bit AWQ raises the mean spread of the 0.5B model from 14.0 to 21.3 points, while at 1.5B the two are similar (6.8 vs. 5.4).

5 Discussion

What is robust? Three findings hold across our models. First, models of 1.2B parameters or fewer are highly format-sensitive: single edits produce spreads of 7–28 points and flip 12–20% of predictions. For these models, the choice of format is as large a source of variance as many reported differences between models or methods. Second, ALL-CAPS is the most harmful common formatting choice. It hurts classification in small QWEN2.5 models and lowers IFEVAL compliance in every model we measured, from 0.5B to 14.7B. Third, trailing whitespace is nearly harmless in chat-formatted prompts: it does not change verbosity and rarely changes predictions (1.6% flips).

Where did the hypotheses go wrong? The hypotheses overestimated the effects of trailing whitespace (H2) and JSON prompts (H4), and underestimated the ALL-CAPS penalty for large models (H3). They also assumed a clean log-linear trend (H5), whereas we observe a steep drop followed by a plateau. One reason for the gap is that much folk knowledge about prompt formatting comes

from raw completion prompting. Chat templates change which edits can matter. Whitespace before an end-of-turn token has little effect, and some templates delete it entirely.

Mechanisms behind large effects. Our error analysis points to a few recurring mechanisms rather than a general "confusion":

- **Label-vocabulary drift.** Under some formats, models answer with off-list labels ("politics", "sport", "news"). This drives the large AG NEWS spreads even at 7–9B.
- **Degenerate abstention.** qwen3b-awq answers "NONE" for 97% of CONLL sentences with plain-list output, but extracts entities when asked for JSON.
- **Over-structuring.** With JSON output on SQUAD, QWEN2.5-3B and QWEN2.5-7B return longer, paraphrased spans, which lowers F1.
- **JSON input disrupting NER.** The LLAMA-3.X models lose 13–17 F1 on CONLL when the sentence is a JSON string value.

Practical recommendations. For models of about 3B parameters or fewer, evaluate over several formats and report the spread, in line with multi-prompt evaluation Mizrahi et al. [2024]. Because edit effects are idiosyncratic and even change under quantization, a format tuned for one checkpoint should not be assumed to transfer to another. At any size, avoid ALL-CAPS templates. Do not assume that JSON helps extraction: wrapping the input in JSON is rarely beneficial, and the output format should be chosen and validated per task. Finally, when measuring effects on long generations, measure the noise floor of the inference stack itself. For IFEVAL, differences of about 3 points can arise from batch-dependent numerics alone.

Alternative explanations. We considered four. (i) *Parser artifacts*: a single lenient parser is shared across conditions, and parse rates are reported (87–100%). Two parser gaps were found by inspecting failures and fixed before final scoring for all conditions alike: the "sport" synonym, and string-valued or list-of-objects JSON. (ii) *Item noise*: addressed with paired tests and permutation nulls. (iii) *Inference nondeterminism*: measured directly (table 7). (iv) *Ceiling effects*: base accuracy rises with scale, which partly confounds the scale trend.

5.1 Limitations

- **Scale range and completeness.** The largest model has 14.7B parameters and is 4-bit quantized. GEMMA-2-9B lacks IFEVAL results, so the $\geq$7B part of H3 rests on three models (qwen7b-awq, llama8b-awq, qwen14b-awq).
- **Model coverage.** We study three open-weight families (seven QWEN2.5 runs, two LLAMA-3.X, one GEMMA-2) and no API models. Hosted APIs may normalize whitespace on the server side.
- **Quantization.** All models of 3B or more are AWQ 4-bit. Our bf16 controls show that quantization changes which edits matter, so results for 16-bit large models may differ in detail.
- **One base template, single edits.** Effects are measured relative to one base template per task. A different anchor could change the deltas, and we do not study interactions between edits.
- **Sample size and multiplicity.** With $n = 300$–500, CIs on classification deltas are about ± 4 points. IFEVAL effects near 3 points sit at the inference noise floor. Holm correction is applied within each classification model$\times$task. IFEVAL and extraction p-values are uncorrected and should be read together with the CIs.
- **Contamination.** All benchmarks are public and are probably in the training data. This affects absolute accuracy more than the within-item format contrasts we study.

Broader impact. This work aims to make LLM evaluation and deployment more reliable by showing where formatting choices can silently change results. We see no direct potential for misuse. One indirect risk is that format sensitivity could be used to cherry-pick formats that favor a particular model. Reporting results over several formats, as we recommend, guards against this.

6 Conclusion

We studied how atomic, meaning-preserving surface edits to chat-formatted prompts change the outputs of ten open-weight LLMs from 0.5B to 14.7B parameters. Surface formatting changes model behavior well beyond the noise floor, but not in the uniform way common beliefs suggest.

The effect is very large below about 1.5B parameters (7–28 point spreads) and levels off at a small but nonzero level (about 4 points) from 7B to 14B. ALL-CAPS is the dominant harmful edit for small models and still costs 3–4 points of instruction compliance at 7–14B. Trailing whitespace does not lengthen outputs, and JSON-structured prompts do not give a uniform extraction gain. Their effects depend strongly on the task and on whether JSON is used for the input or the output. Which edits help or hurt is largely model-specific, and even quantization reshuffles it.

Key takeaway: formatting is part of the experiment. Report results over several formats for small models, avoid non-canonical casing at any size, and validate structured formats per task.

Future work should (i) complete IFEVAL for GEMMA-2-9B and add models of 30B or more to test whether the casing penalty disappears, (ii) repeat the length experiment in raw completion mode, where trailing whitespace changes tokenization at the start of generation, (iii) randomize the base template to separate edit effects from the choice of anchor, (iv) test API models and whether providers normalize whitespace, and (v) use batch-invariant inference kernels He and Thinking Machines Lab [2025] to remove the noise floor on long generations.

References

Berk Atil, Sarp Aykent, Alexa Chittams, Lisheng Fu, Rebecca J. Passonneau, Evan Radcliffe, Guru Rajan Rajagopal, Adam Sloan, Tomasz Tudrej, Ferhan Ture, Zhe Wu, Lixinyu Xu, and Breck Baldwin. Non-determinism of "deterministic" LLM settings. *arXiv preprint arXiv:2408.04667*, 2024.

Stella Biderman, Hailey Schoelkopf, Lintang Sutawika, Leo Gao, Jonathan Tow, Baber Abbasi, et al. Lessons from the trenches on reproducible evaluation of language models. *arXiv preprint arXiv:2405.14782*, 2024.

Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. BoolQ: Exploring the surprising difficulty of natural yes/no questions. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 2924–2936, 2019.

Gemma Team. Gemma 2: Improving open language models at a practical size. *arXiv preprint arXiv:2408.00118*, 2024.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

Horace He and Thinking Machines Lab. Defeating nondeterminism in LLM inference. Thinking Machines Lab: Connectionism, 2025. `https://thinkingmachines.ai/blog/defeating-nondeterminism-in-llm-inference/`.

Jia He, Mukund Rungta, David Koleczek, Arshdeep Sekhon, Franklin X. Wang, and Sadid Hasan. Does prompt formatting have any impact on LLM performance? *arXiv preprint arXiv:2411.10541*, 2024.

Woosuk Kwon, Zhuohan Li, Siyuan Zhuang, Ying Sheng, Lianmin Zheng, Cody Hao Yu, Joseph E. Gonzalez, Hao Zhang, and Ion Stoica. Efficient memory management for large language model serving with PagedAttention. In *Proceedings of the 29th Symposium on Operating Systems Principles (SOSP)*, 2023.

Ji Lin, Jiaming Tang, Haotian Tang, Shang Yang, Wei-Ming Chen, Wei-Chen Wang, Guangxuan Xiao, Xingyu Dang, Chuang Gan, and Song Han. AWQ: Activation-aware weight quantization for on-device LLM compression and acceleration. In *Proceedings of Machine Learning and Systems (MLSys)*, 2024.

Yao Lu, Max Bartolo, Alastair Moore, Sebastian Riedel, and Pontus Stenetorp. Fantastically ordered prompts and where to find them: Overcoming few-shot prompt order sensitivity. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 8086–8098, 2022.

Moran Mizrahi, Guy Kaplan, Dan Malkin, Rotem Dror, Dafna Shahaf, and Gabriel Stanovsky. State of what art? a call for multi-prompt LLM evaluation. *Transactions of the Association for Computational Linguistics*, 12:933–949, 2024.

Felipe Maia Polo, Ronald Xu, Lucas Weber, Mírian Silva, Onkar Bhardwaj, Leshem Choshen, Allysson Flavio Melo de Oliveira, Yuekai Sun, and Mikhail Yurochkin. Efficient multi-prompt evaluation of LLMs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2024.

Pranav Rajpurkar, Jian Zhang, Konstantin Lopyrev, and Percy Liang. SQuAD: 100,000+ questions for machine comprehension of text. In *Proceedings of the 2016 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 2383–2392, 2016.

Abel Salinas and Fred Morstatter. The butterfly effect of altering prompts: How small changes and jailbreaks affect large language model performance. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

Elvis Saravia, Hsien-Chi Toby Liu, Yen-Hao Huang, Junlin Wu, and Yi-Shin Chen. CARER: Contextualized affect representations for emotion recognition. In *Proceedings of the 2018 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 3687–3697, 2018.

Melanie Sclar, Yejin Choi, Yulia Tsvetkov, and Alane Suhr. Quantifying language models' sensitivity to spurious features in prompt design or: How i learned to start worrying about prompt formatting. In *International Conference on Learning Representations (ICLR)*, 2024.

Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D. Manning, Andrew Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, pages 1631–1642, 2013.

Zhi Rui Tam, Cheng-Kuang Wu, Yi-Lin Tsai, Chieh-Yen Lin, Hung-yi Lee, and Yun-Nung Chen. Let me speak freely? a study on the impact of format restrictions on performance of large language models. In *Proceedings of the 2024 Conference on Empirical Methods in Natural Language Processing: Industry Track*, 2024.

Erik F. Tjong Kim Sang and Fien De Meulder. Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition. In *Proceedings of the Seventh Conference on Natural Language Learning (CoNLL)*, pages 142–147, 2003.

Anton Voronov, Lena Wolf, and Max Ryabinin. Mind your format: Towards consistent evaluation of in-context learning improvements. In *Findings of the Association for Computational Linguistics: ACL 2024*, 2024.

Albert Webson and Ellie Pavlick. Do prompt-based models really understand the meaning of their prompts? In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics (NAACL)*, pages 2300–2344, 2022.

Xiang Zhang, Junbo Zhao, and Yann LeCun. Character-level convolutional networks for text classification. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.

Zihao Zhao, Eric Wallace, Shi Feng, Dan Klein, and Sameer Singh. Calibrate before use: Improving few-shot performance of language models. In *International Conference on Machine Learning (ICML)*, pages 12697–12706, 2021.

Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models. *arXiv preprint arXiv:2311.07911*, 2023.

Kaijie Zhu, Jindong Wang, Jiaheng Zhou, Zichen Wang, Hao Chen, Yidong Wang, Linyi Yang, Wei Ye, Yue Zhang, Neil Zhenqiang Gong, and Xing Xie. PromptBench: Towards evaluating the robustness of large language models on adversarial prompts. *arXiv preprint arXiv:2306.04528*, 2023.

A Prompt Examples

All prompts are sent as a single user turn in each model's default chat template. Below are the AG NEWS classification template under `base` and `uppercase`, and the CONLL extraction prompt under the four input/output format cells. Line breaks are shown as in the prompt.

```
Classify the topic of the following news article. The possible topics are world, sports, business, and
technology.

Article: {input}

Answer with one word from the list of topics.
Topic:
```

Example 1: AG NEWS, `base`.

```
CLASSIFY THE TOPIC OF THE FOLLOWING NEWS ARTICLE. THE POSSIBLE TOPICS ARE WORLD, SPORTS, BUSINESS, AND
TECHNOLOGY.

ARTICLE: {input}

ANSWER WITH ONE WORD FROM THE LIST OF TOPICS.
TOPIC:
```

Example 2: AG NEWS, `uppercase` (the inserted article is unchanged).

```
Extract all named entities from the sentence. Entity types: PER (person), ORG (organization), LOC (
location), MISC (other names such as nationalities, events, or products).

Sentence: {input}

List each entity on its own line in the format TYPE: entity text. If there are no entities, write NONE.
[JSON output:] Respond with only a JSON object of the form {"PER": [...], "ORG": [...], "LOC": [...], "
MISC": [...]} listing the entity texts of each type (use empty lists when there are none).
```

Example 3: CONLL, plain input with plain output (JSON output replaces the last line).

```
{
  "task": "Extract all named entities from the sentence. Entity types: PER (person), ORG (organization)
, LOC (location), MISC (other names such as nationalities, events, or products).",
  "sentence": "{input}",
  "output_format": "List each entity on its own line in the format TYPE: entity text. If there are no
entities, write NONE."
}
```

Example 4: CONLL, JSON input with plain output.

B Full Extraction Results

C Reproducibility Details

Software. Python 3.12.12, vLLM 0.32.0, PyTorch 2.13.0 (CUDA 13.0), transformers 5.19.0, pandas 3.0.6, scipy 1.18.1, statsmodels 0.15.0.

Inference. Maximum context length is 2,048–2,560 tokens (the longest prompt plus generation is 1,605 tokens). We use eager execution and enable prefix caching. llama8b-awq runs with tensor parallelism 2. The Llama-3.2-1B checkpoint is an ungated mirror of the official weights (`unsloth/Llama-3.2-1B-Instruct`).

Seeds. Dataset sampling uses seed 42, vLLM uses seed 42, and bootstrap and permutation tests use a NumPy generator seeded with 42. Greedy decoding is still subject to batch-dependent numerics (table 7).

Compute. There was no API spend. Each model took about 4–50 minutes of GPU time for its 24,046 prompts, on 4–12 GB slices of shared 48 GB A6000 GPUs. Because other jobs held 40–45 GB per card, we reserve GPU memory as soon as it becomes free and hand it to vLLM just before it measures free memory. Without this step, about 90% of launches failed to obtain memory.

Table 8: Extraction F1 ($\times 100$) in each input/output format cell. p/p = plain input, plain output; json-in = JSON input, plain output; json-out = plain input, JSON output; json/json = JSON input and output. Best cell per model and task in **bold**.

	SQuAD				CoNLL			
Model	p/p	json-in	json-out	json/json	p/p	json-in	json-out	json/json
qwen0.5b	59.7	57.2	**70.9**	64.6	**17.1**	11.9	16.1	10.2
qwen0.5b-awq	49.9	41.0	**59.9**	41.9	10.7	8.8	**15.1**	12.2
llama1b	61.6	53.7	**69.1**	65.7	21.5	6.1	**21.6**	9.9
qwen1.5b	82.4	81.8	**84.5**	81.8	26.6	28.2	**38.0**	32.5
qwen1.5b-awq	81.4	81.3	**81.8**	78.4	26.5	**28.3**	25.9	26.6
qwen3b-awq	**80.6**	79.8	73.2	70.0	1.4	6.4	**27.1**	**27.1**
qwen7b-awq	82.2	**85.1**	68.0	76.1	27.8	24.7	**52.6**	48.8
llama8b-awq	64.8	**75.4**	58.8	65.2	40.3	13.2	**52.6**	45.0
gemma9b-awq	81.4	81.6	79.3	**83.6**	51.4	50.9	50.9	**53.1**
qwen14b-awq	87.2	87.4	**88.1**	87.0	44.8	36.8	**51.2**	50.3

Artifacts. We save all raw generations, scored outputs, exact prompts (24,046 records per model), and one chat-templated rendering per model, so every number in this paper can be audited.